

Urinalysis



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KEYWORDS

- Urinalysis • Urine sediment examination • Urine chemistries • Urine specific gravity
- Urine dipstick

KEY POINTS

- Urinalysis is a useful laboratory test in documenting urinary tract diseases, and it can also provide information about other systemic diseases, such as liver failure and hemolysis.
- The collection method and storage time and conditions are the most important preanalytical sample variables.
- Preanalytical patient variables include physiologic variables or introduced variables related to treatment or diagnosis.

Urinalysis is an important laboratory test that can be readily performed in veterinary practice and is considered part of a minimum database. It is useful in documenting various types of urinary tract diseases and may provide information about other systemic diseases, such as liver failure and hemolysis. Preanalytical sample variables and patient variables other than those related to disease may influence urinalysis results. Excellent resources are available for more comprehensive information about urinalysis in small animals.^{1–3} Urine may be collected by cystocentesis, urethral catheterization, or voiding and should be evaluated within 30 minutes. If this is not possible, then it may be refrigerated for up to 24 hours or submitted to an outside diagnostic laboratory; however, this may result in crystal precipitation. Refrigeration does not alter urine pH or specific gravity.

IN-HOUSE VERSUS SEND-OUT TESTING

Most veterinary practices can and should do urinalysis in-house, from the standpoint of practice economics and quality of care. The test requires only basic laboratory supplies, including disposable supplies such as a specimen container, disposable pipettes, conical centrifuge tubes, urine dipsticks, glass slides and coverslips, and sediment stain, and equipment, including a centrifuge, a refractometer (preferably

The authors have nothing to disclose.

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Vet Clin Small Anim 45 (2015) 621–637

<http://dx.doi.org/10.1016/j.cvs.2015.02.001>

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a temperature-compensated veterinary model with a feline-specific scale for specific gravity), a microscope, and optionally an automated dipstick reader.⁴⁻⁶ It can be performed easily by clinic technical staff if they are properly trained, making it an inexpensive, technically feasible test to perform. In-house testing is preferred because of faster turnaround time and greater accuracy of results, because delayed analysis is a potential source of introduced error. Another advantage of in-house testing is that results can be correlated more easily with the rest of the patient's clinical picture.

Performing the Complete Urinalysis

A description of performing a urinalysis is provided. The sample should be identified on the sample container using the patient's clinic number or name and the date and should be matched to the reporting form. The reporting form should include patient identification, date, method and time of collection, and any current or recent medications or diagnostic agents. Transfer 0.5 to 1.0 mL of the sample using sterile technique to a sterile tube after mixing the sample and submit or store for aerobic bacteriologic culture. Record the time the urinalysis is performed and whether the sample was refrigerated on the reporting form. If the sample has been refrigerated, allow it to reach room temperature or warm it with mixing to either known patient body temperature or 38°C (101°F). Gently mix the sample by inverting it several times and transfer 3 to 5 mL to a clear conical centrifuge tube and record the color (yellow, brown, black, red, white), clarity (clear, cloudy, turbid, flocculent), and odor (normal or abnormal and describe if possible) on the reporting form. Perform semiquantitative biochemical analysis by immersing the urine dipstick so that all reagent pads are covered with urine, start timing the reactions, and drag edge of strip against rim of tube to remove excess urine. Perform semiquantitative biochemical analysis using a urine dipstick or automated dipstick reader following manufacturer instructions. Record results on reporting form. If the urine sample is grossly discolored (eg, gross hematuria) or turbid, the pigment discolors the reagent pads on the dipstick, making it difficult to read and giving erroneous interpretation. Attempt to clear the urine by centrifuging the sample first. If centrifugation results in a red sediment at the bottom of the conical tube and a clear supernatant, then the dipstick semiquantitative chemical analysis may be performed on the supernatant after transferring it to another tube using a pipette or by decanting. If the urine sample does not clear with centrifugation, then interpret results cautiously. Record results for protein, pH, glucose, ketones, and bilirubin on reporting form; results for specific gravity, urobilinogen, leukocytes, and nitrite are unreliable, inferior to other tests performed as part of the urinalysis, or of no clinical significance and should not be reported. Using an appropriate refractometer measure the specific gravity of the urine sample and report on form. If the sample is discolored or turbid, then the specific gravity can be measured on the supernatant after centrifugation. Centrifuge 3 to 6 mL of the sample in the conical tube for 5 minutes at 1500 to 2000 rpm (450 g). Do not use the brake to stop the centrifuge as this may result in suspension of the sediment. Transfer all but 0.5 to 1.0 mL of the supernatant depending on the volume of urine in the tube to another tube by using a pipette or by decanting. If the sample cleared with centrifugation, then dipstick semiquantitative chemical analysis and specific gravity determination may be performed as mentioned previously. Suspend the remaining sediment pellet in the 0.5 to 1.0 mL of supernatant by tapping the conical tip of the tube gently on the table top. Use less urine to suspend the sediment pellet if less urine was available for centrifugation. Transfer a drop of the suspension to a clean glass microscope slide using a pipette and place a glass coverslip over it. There should be enough fluid to

fill the area under the coverslip, but the coverslip should not float. Examine the slide immediately using bright field light microscopy optimized for contrast by closing the condenser diaphragm if the microscope has one or by lowering the condenser. Using the 10× objective (100× magnification) scan at least 2 edges of the area under the coverslip and note presence of casts and crystals as amount per low power field (#/lpf). Using the 40× objective (400× magnification) scan at least 10 fields of the area under the coverslip for cells (red blood cells [RBCs], white blood cells [WBCs], epithelial cells), amorphous crystalline material, microorganisms (bacteria, yeast, parasites), spermatozoa, and fat droplets and record as amount per high power field (#/hpf). Estimate numbers if it is difficult to quantitate, and if too numerous to count record as such. A second slide may be prepared as described, air dried, stained with a modified Wright stain, Wright-Giemsa stain, or Gram stain, and examined using the 10× and 40× objectives as described and with immersion oil at 100× (1000× magnification) for identification of bacteria.^{7–10}

STAINED URINE SEDIMENT EXAMINATION

Preanalytical Sample Variables

A urine sample submitted for analysis should be free of any contaminants, collected into a clean, opaque, airtight, sterile container, and analyzed within 60 minutes of collection; however, as this is not always the case, the clinician should consider the influence of preanalytical sample (ie, nonbiologic) variables when interpreting urinalysis results. The most important of these variables are collection method and storage time and conditions. The collection method is relevant with regard to potential contamination. For example, samples collected by cystocentesis typically have a small amount of iatrogenic blood contamination, voided samples often contain contaminants from the lower urogenital tract (eg, bacteria, epithelial cells, spermatozoa, blood) or the environment (eg, bacteria, plant material, debris), samples obtained by urinary catheterization may contain lower urogenital tract contaminants or iatrogenic hemorrhage, and samples obtained from the floor or examination table or from a litter box lined with nonabsorbent material are often contaminated with microbes and debris. Storage time and conditions may affect urinalysis results. For example, refrigeration may promote crystal formation; prolonged storage of samples at room temperature may result in degeneration of cells and casts, altered crystal formation, or bacterial overgrowth and resultant secondary artifacts (eg, altered pH, decreased glucose concentration); and exposure to light or air may alter the composition of the sample.

Preanalytical Patient Variables

Preanalytical patient variables include physiologic variables or introduced variables related to treatment or diagnosis. Physiologic variables that affect results include diet, time of day, and reproductive factors. For example, diet may influence urine pH or crystal formation, samples collected first thing in the morning tend to be more concentrated and may have altered cellular morphology or microbial viability, and estrus or recent breeding may influence microscopic findings. Introduced patient variables include administration of drugs, fluids, or other exogenous agents. For example, corticosteroids and diuretics interfere with renal concentrating ability, corticosteroids may also cause proteinuria, antimicrobials may mask urinary tract infections or form crystals, hydroxyethyl starch interferes with measurement of specific gravity, and radiographic contrast agents may interfere with measurement of specific gravity or biochemical analytes or may form crystals.

RESULTS OF COMPLETE URINALYSIS

Urine Appearance

Color

Normal urine is typically transparent and yellow or amber on visual inspection. Primarily 2 pigments impart the yellow coloration: urochrome and urobilin. Urochrome is a sulfur-containing oxidation product of the colorless urochromogen. Urobilin is a degradation product of hemoglobin. Because the 24-hour urinary excretion of urochrome is constant, highly concentrated urine is amber in color, whereas dilute urine may be transparent or light yellow color. The intensity of the color is in part related to the volume of urine collected and the concentration of urine produced; therefore, it should be interpreted in the context of the urine specific gravity. Caution must be used not to overinterpret the significance of urine color as part of a complete urinalysis. Significant disease may exist when urine is normal in color. Abnormal urine color may be caused by the presence of several endogenous or exogenous pigments. Although the abnormal color indicates a problem, it provides nonspecific information. Causes of abnormal coloration should be investigated with appropriate laboratory tests and examination of urine sediment. Detection of abnormal urine color should prompt questions related to diet, administration of medication, environment, and collection technique. Knowledge of urine color may also be important in interpreting colorimetric test results because it may induce interference with the test.

Urine color that is anything other than yellow or amber is abnormal. There are many potential causes of discolored urine ([Fig. 1](#), [Table 1](#)). The most common abnormal urine color in dogs and cats is red, brown, or black, which may be caused by hematuria, hemoglobinuria, myoglobinuria, and bilirubinuria.

Pale yellow urine Urine that is pale yellow or clear in appearance may be normal or may indicate a polyuric state. Urine may be appropriately dilute if it is associated with recent consumption or administration of fluids, consumption of a diet containing low quantities of protein or high quantities of sodium chloride, glucocorticoid excess, or administration of diuretics. Urine would be considered to be inappropriately

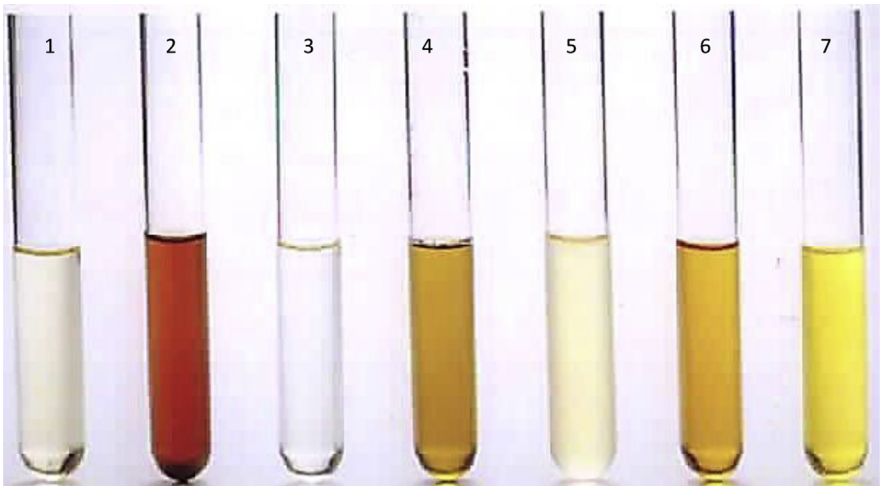


Fig. 1. Urine samples having various colors from clear (3) to pale yellow (1, 5), dark yellow (4, 6, 7), and red due to hematuria (2).

Table 1
Potential causes of discolored urine

Urine Color	Causes	Urine Color	Causes
Yellow or amber	Urochromes Urobilin	Yellow-brown or green-brown	Bile pigments
Deep yellow	Highly concentrated urine Quinacrine ^a Nitrofurantoin ^a Phenacetin ^a Riboflavin (large quantities) ^a Phenolsulfonphthalein (acidic urine) ^a	Brown to black (brown or red-brown when viewed in bright light in thin layer)	Melanin Methemoglobin Myoglobin Bile pigments Thymol ^a Phenolic compounds ^a Nitrofurantoin ^a Nitrites ^a Naphthalene ^a Chlorinated hydrocarbons ^a Aniline dyes ^a Homogentisic acid ^a

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Table 1
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Urine Color	Causes	Urine Color	Causes
Blue	Methylene blue Indigo carmine and indigo blue dye ^a Indicans ^a Pseudomonas infection ^a Water-soluble chlorophyll ^a Rhubarb ^a Toluidine blue ^a Triamterene ^a Amitriptyline ^a Anthraquinone ^a Blue food dye ^a	Colorless	Very dilute urine (diuretics, diabetes mellitus, diabetes insipidus, glucocorticoid excess, fluid therapy, overhydration)
Green	Methylene blue Dithiazanine Urate crystalluria Indigo blue ^a Evan blue ^a Bilirubin Biliverdin Riboflavin ^a Thymol ^a Phenol ^a Triamterene ^a Amitriptyline ^a Anthraquinone ^a Green food dye ^a	Milky white	Lipid Pyuria Crystals

Red, pink, red-brown, red-orange, or orange	Hematuria Hemoglobinuria Myoglobinuria Porphyrinuria Congo red Phenolsulfonphthalein (following alkalization) Neoprontosil Warfarin (orange) ^a Food pigments (rhubarb, beets, blackberries) ^a Carbon tetrachloride ^a Phenazopyridine Phenothiazine ^a Diphenylhydantoin ^a Bromsulphalein (following alkalization) Chronic heavy metal poisoning (lead, mercury) ^a Rifampin ^a Emodin ^a Phenindione ^a Eosin ^a Rifabutin ^a Acetazolamide ^a Red food dye ^a	Brown	Methemoglobin Melanin Sulfasalazine ^a Nitrofurantoin ^a Phenacetin ^a Naphthalene ^a Sulfonamides ^a Bismuth ^a Mercury ^a Feces (rectal-urinary fistula) Fava beans ^a Rhubarb ^a Sorbitol ^a Metronidazole ^a Methocarbamol ^a Anthracin cathartics ^a Clofazimine ^a Primaquine ^a Chloroquine ^a Furazolidone ^a Copper toxicity
Orange-yellow	Highly concentrated urine Excess urobilin Bilirubin Phenazopyridine Sulfasalazine ^a Fluorescein sodium ^a Flutamide ^a Quinacrine ^a Phenacetin ^a 2,4-Dichlorophenoxyacetic acid ^a Acetazolamide ^a Orange food dye ^a		

^a Only observed in human beings.

concentrated if it were dilute in the presence of dehydration. Diseases that may be associated with persistently dilute urine include diabetes mellitus, diabetes insipidus, hyperadrenocorticism, hypoadrenocorticism, hypercalcemia, hyperthyroidism, and renal failure. If urine is pale yellow or clear, the urine specific gravity is often less than 1.015. A simple test to determine whether polyuria is persistent is to determine the urine specific gravity of a sample collected in the morning. Other tests should include serum biochemical analysis and a complete urinalysis. Additional testing may include measurement of serum thyroxine concentration, adrenal function testing, or monitoring urine specific gravity after several days of vasopressin administration.

Red, brown, or black urine Red, brown, or black urine suggests the presence of blood, hemoglobin, myoglobin, or bilirubin. A positive occult blood reaction is obtained when urine contains any of these substances. Discoloration of urine may also result in false-positive reactions on other urine dipstick test pads. Analysis of urine sediment reveals the presence of RBCs if the discoloration is due to hematuria. If no RBCs are present on microscopic examination of urine sediment, presence of hemoglobin, myoglobin, or bilirubin should be suspected. Examination of plasma color may aid in differentiating these. If the urine is discolored because of myoglobin, the plasma is clear because myoglobin in plasma is not bound significantly to a carrying protein, which results in filtration and excretion of myoglobin. If the plasma is pink, it is suggestive of hemoglobin. If the plasma is yellow, it is suggestive of bilirubin; serum bilirubin concentration should also be increased. Myoglobinuria indicates muscle damage; serum creatine kinase activity is often increased in this setting. Hemoglobinemia indicates intravascular hemolysis resulting from immune-mediated, parasite-mediated, or drug-mediated destruction of RBCs. Hyperbilirubinemia may result from liver disease, post-hepatic obstruction, or hemolysis.

Milky white urine Milky white urine may be due to the presence of WBCs (pyuria), lipid, or crystals. The more concentrated the urine sample is the more opaque it may appear. The presence of pyuria secondary to a bacterial urinary tract infection is the most common cause of milky white urine; however, pyuria may occur because of inflammation and not be associated with an infection. Lipiduria may be observed in healthy animals but is frequently observed in cats affected with hepatic lipidosis. Crystalluria if heavy and present in a concentrated urine sample may also result in milky white urine. Microscopic examination of urine sediment aids in differentiation of these causes.

Clarity

Urine is typically clear but may become less transparent with pigmenturia, crystalluria, hematuria, pyuria, lipiduria, or when other compounds such as mucus are present. Depending on the cause, increased turbidity may disappear with centrifugation of the sample.

Odor

Normal urine has a slight odor of ammonia; however, the odor depends on urine concentration. Some species, such as cats and goats, have pungent urine odor because of urine composition. Bacterial infection may result in a strong odor due to pyuria; a strong ammonia odor may occur if the bacteria produce urease.

URINE CHEMICAL ANALYSIS

Urine must be at room temperature for accurate measurement of specific gravity and for chemical analysis. These tests are usually done before centrifugation; however, if

urine is discolored or turbid, it may be beneficial to perform these tests on the supernatant.

Specific Gravity

Specific gravity is defined as the ratio of the weight of a volume of liquid to the weight of an equal volume of distilled water; therefore, it depends on the number, size, and weight of particles in the liquid. It is different from osmolality, which depends only on the number of particles in the liquid; measurement of osmolality requires specialized instrumentation. Urine specific gravity is determined using a refractometer designed for veterinary samples, which includes a scale calibrated specifically for cat urine. Urine specific gravity for species other than cats should be determined using the scale for dogs. In healthy animals, urine specific gravity is highly variable, depending on fluid and electrolyte balance of the body. Interpretation of urine specific gravity, therefore, depends on the clinical presentation and findings of serum chemical analysis. An animal that is dehydrated or has other causes of prerenal azotemia has hypersthenuric urine with a specific gravity of 1.025 to 1.040 (depending on the species). Dilute urine in a dehydrated or azotemic animal is abnormal and could be caused by renal failure, hypoadrenocorticism or hyperadrenocorticism, hypercalcemia, diabetes mellitus, hyperthyroidism, diuretic therapy, or diabetes insipidus. Glucosuria increases the urine specific gravity despite increased urine volume.

Semiquantitative, Colorimetric Reagent Strips

Reagent strips such as Multistix (Siemens Healthcare, Erlangen, Germany) or Chem-strip (Roche Diagnostics, Indianapolis, IN) can be used to perform several semiquantitative chemical evaluations simultaneously. They are used routinely to determine urine pH, protein, glucose, ketones, bilirubin/urobilinogen, and occult blood. Some reagent strips include test pads for leukocyte esterase (for detection of WBCs), nitrite (for detection of bacteria), and (urine specific gravity) USG; these are not valid in animals and should not be used. Reagent strips are adversely affected by moisture and have a limited shelf life. Bottles should be kept tightly capped, and unused strips should be discarded after their expiration date.

pH

Urine pH is typically acidic in dogs and cats and alkaline in horses and ruminants but varies depending on diet, medications, or presence of disease. Reagent strip colorimetric test pads for pH determination are accurate to within ± 0.5 pH units.¹¹ For example, a reading of 6.5 means the actual pH is likely to be between 6.0 and 7.0. Portable pH meters are more accurate than pH colorimetric test pads. A bacterial urinary tract infection with a urease-producing microbe results in alkaluria. Urine pH affects crystalluria because some crystals, such as struvite, form in alkaline urine, whereas other crystals, such as cystine, form in acidic urine.

Protein

The protein test pad uses a color indicator (tetrabromophenol blue), which detects primarily albumin in urine. Results range from 10 to 1000 mg/dL. Proteinuria can occur from prerenal (fever, strenuous exercise, seizures, extreme environmental temperature, and hyperproteinemia), renal (primarily glomerular and occasionally tubular disease), or postrenal (inflammation, hemorrhage, and infection) causes. A positive reaction must be interpreted in light of USG, pH, and urine sediment examination. For example, a trace amount of protein in concentrated urine is less significant than a trace amount of protein in dilute urine. Alkaluria gives a false-positive reaction. Likewise, presence of other proteins, such as Bence-Jones proteins, gives false-negative

results. Proteinuria can be measured using sulfosalicylic acid precipitation, which detects albumin and globulins; however, it is not accurate in dogs and cats. If proteinuria is present with an inactive urine sediment, its significance can be verified and quantitated by dividing the urine protein concentration by the urine creatinine concentration (urine protein to urine creatinine ratio; UP:UC). Interpretation of UP:UC is as follows: less than 0.5:1.0 (dogs) and less than 0.4:1.0 (cats) is normal, 0.4 or 0.5 to 1.0:1.0 is questionable, and greater than 1.0:1.0 is abnormal. With primary renal azotemia, UP:UC greater than 0.4:1.0 in cats and 0.5:1.0 in dogs is considered abnormal.¹² A semiquantitative microalbuminuria test is available to detect urinary albumin in the range of 1 to 30 mg/dL. It uses enzyme-linked immunosorbent assay technology specific for canine or feline albumin. Because of minor species differences to albumin, there are different kits for dogs and cats. The microalbuminuria test detects lower concentrations of albumin than a standard dipstick test pad. Hematuria must be macroscopic to increase the microalbuminuria or UP:UC; however, pyuria increases both.

Glucose

Glucose is detected by a glucose oxidase enzymatic reaction that is specific for glucose. Glucosuria is not present normally because the renal threshold for glucose is greater than 180 mg/dL in most species and greater than 240 mg/dL in cats. With euglycemia, the amount of filtered glucose is less than the renal threshold and all the filtered glucose is reabsorbed in the proximal renal tubules. Glucosuria can result from hyperglycemia (due to diabetes mellitus, excessive endogenous or exogenous glucocorticoids, or stress) or from a proximal renal tubular defect (such as primary renal glucosuria or Fanconi syndrome). If glucosuria is present, blood glucose concentration should be determined. False-negative results can occur with high urinary concentrations of ascorbic acid (vitamin C) or with formaldehyde (a metabolite of the urinary antiseptic, methenamine, which may be used for prevention of bacterial urinary tract infections). False-positive results may occur if the sample is contaminated with hydrogen peroxide, chlorine, or hypochlorite (bleach).

Ketones

Ketones are produced from fatty acid metabolism and include acetoacetic acid, acetone, and β -hydroxybutyrate. The ketone test pad detects acetone and acetoacetic acid but not β -hydroxybutyrate. The test pad contains nitroprusside that reacts with acetoacetic acid and acetone to cause a purple color change; it is more sensitive to acetoacetic acid than acetone. Ketonuria is associated with primary ketosis (ruminants), ketosis secondary to diabetes mellitus (small animals), consumption of low-carbohydrate diets (especially in cats), and occasionally prolonged fasting or starvation. A false-positive reaction can occur with the presence of reducing substances in the urine.

Bilirubin/Urobilinogen

When hemoglobin is degraded, the heme portion is converted to bilirubin, which is conjugated in the liver and excreted in bile. Some conjugated bilirubin is filtered by the glomerulus and excreted in urine. The kidney can metabolize hemoglobin to bilirubin and secrete it in dogs but not in cats. Male dogs have a higher secretory ability than female dogs. Dipstick reagent pads use diazonium salts to create a color change and are more sensitive to conjugated bilirubin than unconjugated bilirubin. Bilirubinuria occurs when the level of conjugated bilirubin exceeds the renal threshold as with liver disease or hemolysis. In dogs with concentrated urine, a small amount of bilirubin can be normal. Pigmenturia and phenothiazine may result in a false-positive reaction; false-negative reactions may occur with large amounts of urinary ascorbic acid (vitamin C).

Urobilinogen, formed from bilirubin by intestinal microflora, is absorbed into the portal circulation and is excreted renally. A small amount of urinary urobilinogen is normal. Increased urinary urobilinogen level occurs with hyperbilirubinemia; a negative test result may be observed with biliary obstruction; however, the test is not specific enough to be clinically useful.

Occult blood

The occult blood test pad uses a pseudoperoxidase method to detect intact RBCs, hemoglobin, and myoglobin. A positive reaction can be due to hemorrhage (hematuria), intravascular hemolysis (hemoglobinuria), or myoglobinuria. The last 2 processes can be distinguished by examination of plasma; plasma appears pink to red after intravascular hemolysis, whereas myoglobin is rapidly cleared from plasma, resulting in clear plasma. As with other colorimetric test pads, discolored urine may yield false-positive results. A positive result should be interpreted with microscopic examination of urine sediment.

URINE SEDIMENT

Microscopic examination of urine sediment should be part of a routine urinalysis. For centrifugation, 3 to 5 mL of urine is transferred to a conical centrifuge tube. Urine is centrifuged at 1000 to 1500 rpm for 5 minutes. The supernatant is decanted, leaving 0.5 to 1.0 mL of urine and sediment in the tip of the conical tube. The sediment is resuspended by tapping the tip of the conical tube against the table several times. A few drops of the sediment are transferred to a glass slide, and a coverslip is applied. Examination of unstained urine is recommended for routine samples. Microscopic examination is performed at 10 $\times$ objective (for crystals, casts, and cells) and 40 $\times$ objective (for cells and bacteria) magnifications. Contrast of the sample is enhanced by closing the iris diaphragm and lowering the condenser of the microscope. Stains such as Sedi-Stain and new methylene blue can be used to aid in cell identification but may dilute the specimen and introduce artifacts such as stain precipitate and crystals. Use of a modified Wright stain increases the sensitivity, specificity, and positive and negative predictive values for detection of bacteria.

Red Blood Cells

In an unstained preparation, RBCs are small and round and have a slight orange tint and a smooth appearance (**Fig. 2**). Normal urine should contain less than 5 RBCs per field at 400 $\times$ magnification (40 $\times$ objective). Increased RBC count in urine (hematuria) indicates hemorrhage somewhere in the urogenital system; however, sample collection by cystocentesis or catheterization may induce hemorrhage.

White Blood Cells

WBCs are slightly larger than RBCs and have a grainy cytoplasm (see **Fig. 2**; **Fig. 3**). Normal urine should contain less than 5 WBCs per field at 400 $\times$ magnification (40 $\times$ objective). Increased WBC count (pyuria) can occur because of inflammation, infection, trauma, or neoplasia. Catheterization or collection of voided urine may introduce a few WBCs from the urogenital tract.

Epithelial Cells

Transitional epithelial cells, a common urine contaminant derived from the bladder and proximal urethra, resemble WBCs but are larger. They have a greater amount of grainy cytoplasm and a round, centrally located nucleus. In a voided urine sample, squamous epithelial cells may be observed. They are large, oval to cuboidal, and may or may not



Fig. 2. Red blood cells and 1 white blood cell (arrow).

contain a nucleus. Occasionally, neoplastic transitional cells may be observed in an animal with a transitional cell carcinoma. Neoplastic squamous cells may be observed in an animal with a squamous cell carcinoma.

Cylindruria (Casts)

Casts are elongated, cylindrical structures formed by mucoprotein congealing within renal tubules and may contain cells (**Fig. 4**). Hyaline casts are pure protein precipitates, are transparent, have parallel sides and rounded ends, and are composed of mucoprotein. They may occur with fever, exercise, and renal disease. Epithelial cellular casts form from entrapment of sloughed tubular epithelial cells in the

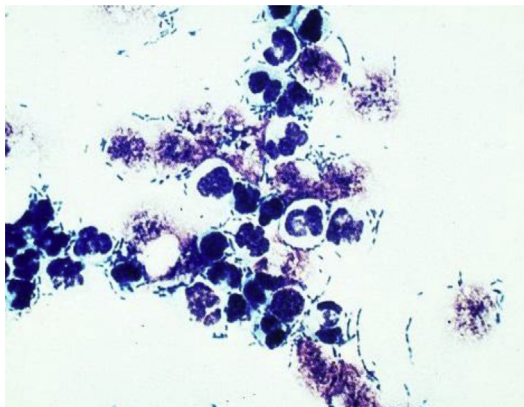


Fig. 3. White blood cells (neutrophils) and bacteria.



Fig. 4. Granular cast.

mucoprotein; they may be observed with renal tubular disease. Granular casts are thought to represent degenerated epithelial cellular casts. Waxy casts have a granular appearance and are thought to arise from degeneration of long-standing granular casts. They typically have sharp borders with broken ends. Other cellular casts include erythrocyte casts and WBC casts and are always abnormal. Erythrocyte casts form because of renal hemorrhage. WBC casts occur because of renal inflammation, as with pyelonephritis. Fatty casts are not common but can be observed with disorders of lipid metabolism, such as diabetes mellitus. A few hyaline or granular casts are considered normal. However, the presence of cellular casts or other casts in high numbers indicates renal damage and may be one of the earliest abnormalities determined by laboratory tests noted with toxic damage to renal epithelial cells (eg, gentamicin, amphotericin B).

Infectious Organisms

The presence of bacteria in urine collected by cystocentesis indicates infection. Small numbers of bacteria from the lower urogenital tract may contaminate voided samples or samples collected by catheterization and do not indicate infection. Bacterial rods are most easily identified in urine sediment (see [Fig. 3](#)). Particles of debris may be mistaken for bacteria. Suspected bacteria can be confirmed by staining urine sediment with Gram stain or modified Wright stain; however, aerobic culture is best to confirm a bacterial urinary tract infection. Rarely, yeast and fungi ([Fig. 5](#)) and parasitic

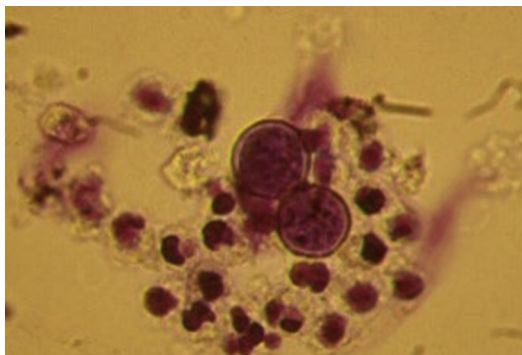


Fig. 5. Fungal organism (*Blastomyces* spp) and white blood cells (neutrophils).

ova may be observed in urine sediment. Their presence is not always associated with clinical disease. Parasitic ova observed include *Stephanus dentatus*, *Capillaria plica*, *Capillaria felis* (Fig. 6), and *Diocetophyma renale*. In addition, microfilariae of *Dirofilaria immitis* may be observed in urine sediment.

Crystals

Many urine sediments contain crystals. The type of crystal present depends on urine pH, concentration of crystallogenic materials, urine temperature, and length of time between urine collection and examination. Crystalluria is not synonymous with urolithiasis and is not necessarily pathologic. Furthermore, uroliths may form without observed crystalluria. Struvite crystals (Fig. 7) are commonly observed in canine and feline urine. Struvite crystalluria in dogs is not a problem unless there is a concurrent bacterial urinary tract infection with a urease-producing microbe. Without an infection, struvite crystals in dogs are not associated with struvite urolith formation. However, struvite uroliths form in some animals (eg, cats) without a bacterial urinary tract infection. In these animals, struvite crystalluria may be pathologic. Struvite crystals appear typically as coffin lids or prisms; however, they may be amorphous. Calcium oxalate crystalluria occurs less commonly in dogs and cats; if persistent, it may indicate an increased risk for calcium oxalate urolith formation. However, calcium oxalate and calcium carbonate crystalluria is common in healthy horses and cattle. Calcium oxalate dihydrate crystals appear as squares with an X in the middle (Fig. 8) or as envelope shaped. Calcium oxalate monohydrate crystals are dumb-bell shaped. An unusual form of calcium oxalate crystals is typically seen in association with ethylene glycol toxicity. These crystals occur in neutral to acidic urine. They are small, flat, and colorless and are shaped like picket fence posts. Ammonium acid urate crystals suggest liver disease (eg, portosystemic shunt). These crystals occur in acidic urine and are yellow-brown spheres with irregular, spiny projections (Fig. 9); however, they may also be amorphous. Certain species, such as birds and reptiles, and certain breeds of dogs, specifically Dalmatians, can normally have ammonium acid urate crystalluria. Cystine crystals are 6 sided and of variable size



Fig. 6. *Capillaria felis* ovum.



Fig. 7. Struvite crystals, red blood cells, and spermatozoon (arrow).

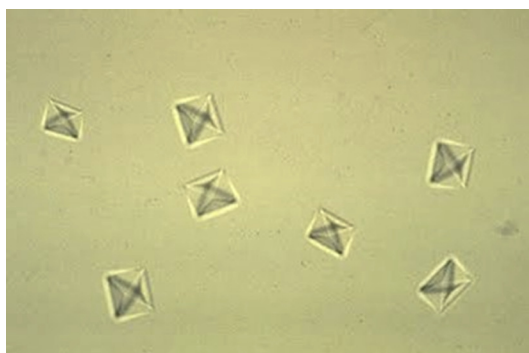


Fig. 8. Calcium oxalate dihydrate crystals.

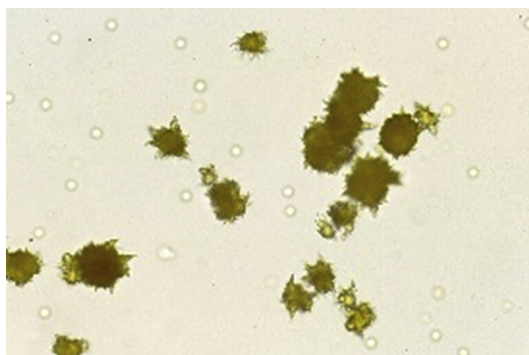


Fig. 9. Ammonium urate crystals.



Fig. 10. Cystine crystals.

(**Fig. 10**). They occur in acidic urine. Presence of cystine crystals represents a proximal tubular defect in amino acid reabsorption. Cystinuria has been reported to occur in many breeds of dogs and rarely in cats. Dachshunds, Newfoundlands, English bulldogs, and Scottish terriers have a high incidence of cystine urolithiasis. Bilirubin crystals occur with bilirubinuria; however, they may be normal in small numbers in dogs.

Lipids

Fat droplets are commonly present in urine from dogs and cats and may be mistaken for RBC. They often vary in size and tend to float on a different plane of focus than the remainder of the sediment. They are not considered to be pathologic.

Spermatozoa

Spermatozoa may be observed normally in urine collected from male dogs (see **Fig. 7**).

Plant Material

Occasionally, plant material may be observed in urine samples collected by voiding (**Fig. 11**). When present, it indicates contamination of the urine sample and is not pathologic.



Fig. 11. Plant pollen in a voided urine sample.

REFERENCES

1. Osborne CA, Stevens JB. Urinalysis: a clinical guide to compassionate patient care. Shawnee Mission (KS): Bayer Corporation; 1999.
2. Wamsley H, Alleman R. Complete urinalysis. In: Elliott J, Grauer GF, editors. BSAVA manual of canine and feline nephrology and urology. Gloucester (United kingdom): British Small Animal Veterinary Association; 2007. p. 87–116.
3. Fry MM. Urinalysis. In: Bartges JW, Polzin DJ, editors. Nephrology and urology of small animals. Ames (IA): Blackwell Publishing; 2011. p. 46–57.
4. Defontis M, Bauer N, Failing K, et al. Automated and visual analysis of commercial urinary dipsticks in dogs, cats and cattle. *Res Vet Sci* 2013;94(3):440–5.
5. Bauer N, Rettig S, Moritz A. Evaluation the Clinitek status automated dipstick analysis device for semiquantitative testing of canine urine. *Res Vet Sci* 2008; 85(3):467–72.
6. George JW. The usefulness and limitations of hand-held refractometers in veterinary laboratory medicine: an historical and technical review. *Vet Clin Pathol* 2001;30(4):201–10.
7. O’Neil E, Horney B, Burton S, et al. Comparison of wet-mount, Wright-Giemsa and Gram-stained urine sediment for predicting bacteriuria in dogs and cats. *Can Vet J* 2013;54(11):1061–6.
8. Swenson CL, Boisvert AM, Gibbons-Burgener SN, et al. Evaluation of modified Wright-staining of dried urinary sediment as a method for accurate detection of bacteriuria in cats. *Vet Clin Pathol* 2011;40(2):256–64.
9. Swenson CL, Boisvert AM, Kruger JM, et al. Evaluation of modified Wright-staining of urine sediment as a method for accurate detection of bacteriuria in dogs. *J Am Vet Med Assoc* 2004;224(8):1282–9.
10. Way LI, Sullivan LA, Johnson V, et al. Comparison of routine urinalysis and urine Gram stain for detection of bacteriuria in dogs. *J Vet Emerg Crit Care (San Antonio)* 2013;23(1):23–8.
11. Heuter KJ, Buffington CA, Chew DJ. Agreement between two methods for measuring urine pH in cats and dogs. *J Am Vet Med Assoc* 1998;213(7):996–8.
12. Littman MP, Daminet S, Grauer GF, et al. Consensus recommendations for the diagnostic investigation of dogs with suspected glomerular disease. *J Vet Intern Med* 2013;27(Suppl 1):S19–26.